DIW Platform — Material Extrusion 3D Printing System



> **Direct Ink Writing (DIW)** 3D printing platform developed at **NUS CDE AML Lab**. This system adapts mainstream 3D printing hardware (MKS-Monster8 + Marlin) for room-temperature material extrusion, enabling precise deposition of functional inks (cement pastes, gels, slurries, etc.).

Table of Contents

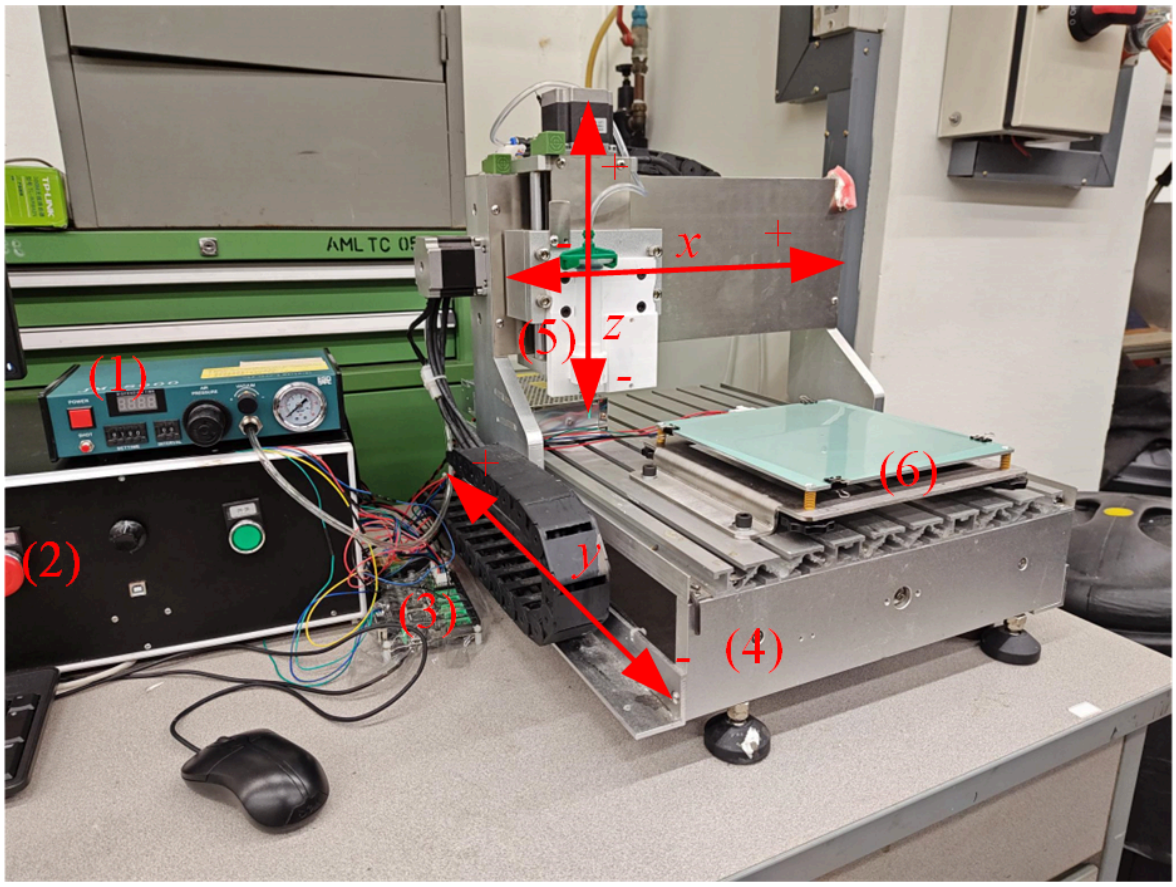
1. Introduction
    - 1.1 System Overview
    - 1.2 Controller Board & Firmware
    - 1.3 Stepper Motors, Drivers & Endstops
    - 1.4 Pressure Dispenser
2. Getting Started
    - 2.1 Prepare G-Code
    - 2.2 Connection & Communication
3. Tips & Troubleshooting
4. Support Materials

---

1. Introduction

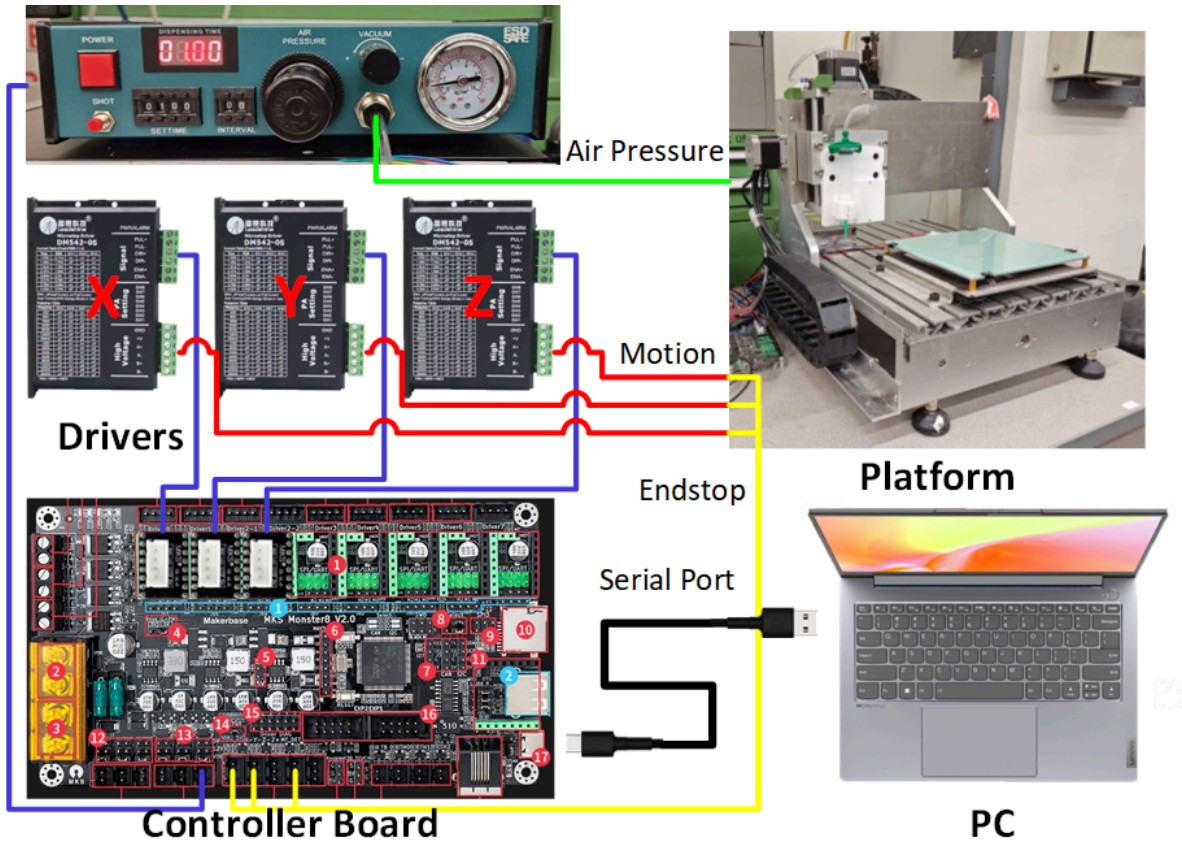
Fixture [2] and assembly [1] are available in the 3Dmodels directory.

Figure 1. Overview of DIW platform: (1) Fluid dispenser, (2) Step motor driver box, (3) Controller board, (4) Motion control platform, (5) Syringe barrel, (6) Substrate.



The platform consists of a controller board, fluid dispenser, and a 3-axis motion platform. The wiring connections between components are illustrated below:

Figure 2. DIW platform wiring diagram.



1.1 System Overview

Component	Specification
Controller	MKS-Monster8 (STM32F407VET6, 168MHz)
Motors	57 stepper motors × 3 axes
Dispenser	JND-983A pneumatic dispensing valve
Nozzles	0.4 mm / 0.8 mm diameter
Firmware	Marlin 2.1.2.4 (customized)
Slicer	PrusaSlicer 2.7.4
Host Software	Pronterface

1.2 Controller Board & Firmware

The controller uses the MKS-Monster8 board from MakerBase, supporting mainstream 3D printing firmware (Marlin, Klipper).

Figure 3. MKS-Monster8 controller board.



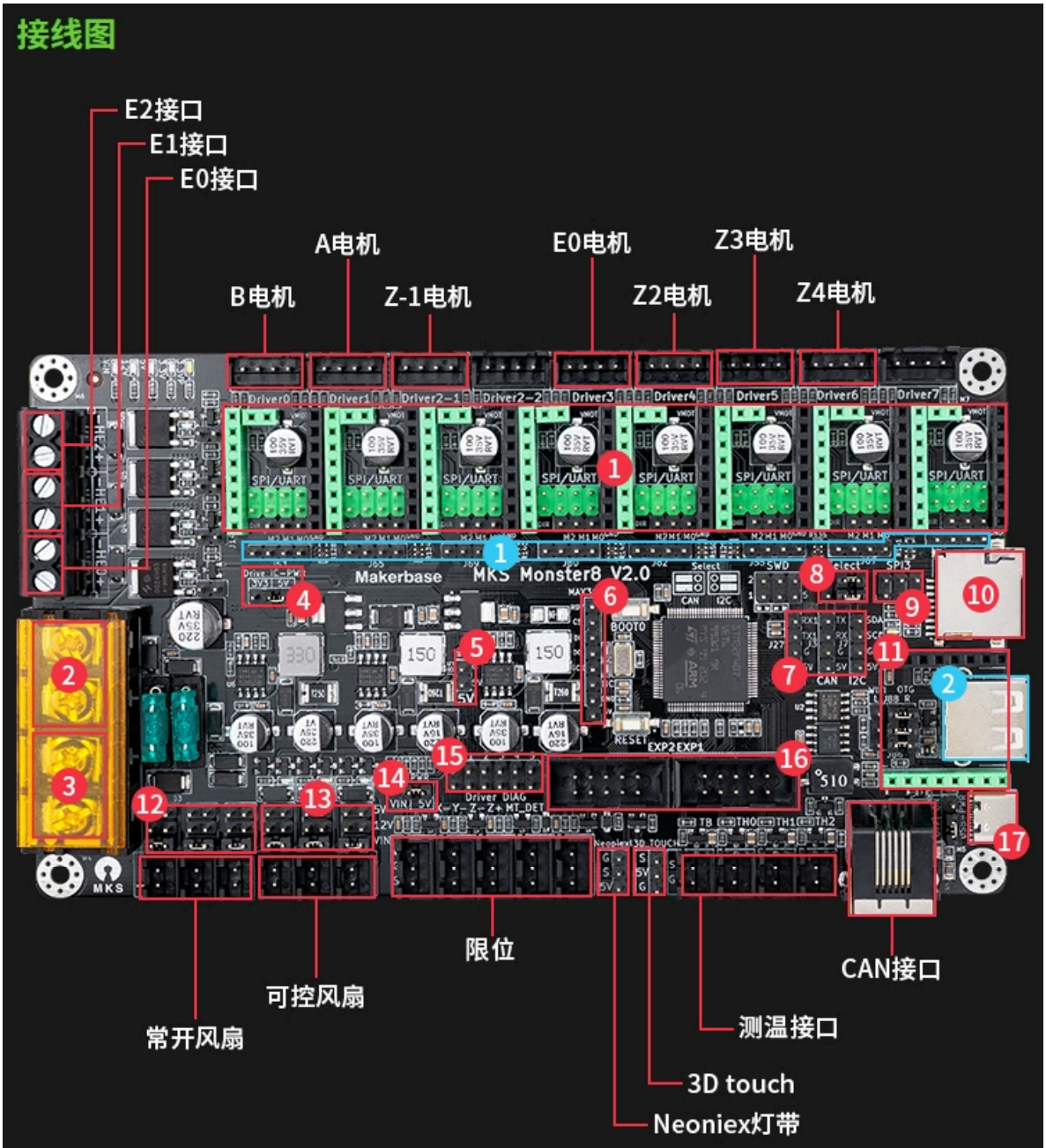
References: GitHub | Video Guide | Marlin Setup | Klipper Setup

Technical Parameters

- **MCU:** STM32F407VET6 (168MHz, 512KB Flash, 192KB RAM)
- **Power Supply:** DC 12-24V input (2× MP1584EN outputs: DC12V for fans, DC5V)
- **Fans:** 3× PWM fans + 3× power outputs (selectable via jumper: VIN/DC12V/DC5V)
- **Motors:** 9 driver interfaces (Driver 0-7)
- **Displays:** EXP1/EXP2 for MKS MINI12864, MKS TS35, LCD12864, LCD2004
- **Serial:** USART1 (PA9/PA10) for MKS Robin or custom serial communication
- **Endstops:** 6 inputs with power selection (X-, X+, Y-, Y+, Z-, Z+) + 3D TOUCH (PA8)
- **Storage:** 4KB EEPROM (I2C), onboard microSD card (SPI)
- **Communication:** CAN transceiver, SPI interface, virtual USB device, UDISK
- **Drivers:** TMC UART/SPI modes, SENSORLESS_HOMING (Diag0-5)
- **Driver Voltage:** Selectable 5V/3.3V
- **Protection:** TVS spike protection, reverse connection protection
- **Boot Mode:** DFU via Boot0 button

Figure 4-5. Pinout and connector layout.

接线图



1 电机驱动器插槽　2 热床　3 电源

4 驱动电压选择　5 5V输出　6 MAX31865

7 I2C/CAN/SUART1扩展端口

8 I2C和CAN选择，默认选择I2C，即EERPOM功能

9 TF卡信号，SPI3扩展口　10 TF卡槽　11 WIFI模块插槽

12 常开风扇电压选择　13 可控风扇电压选择

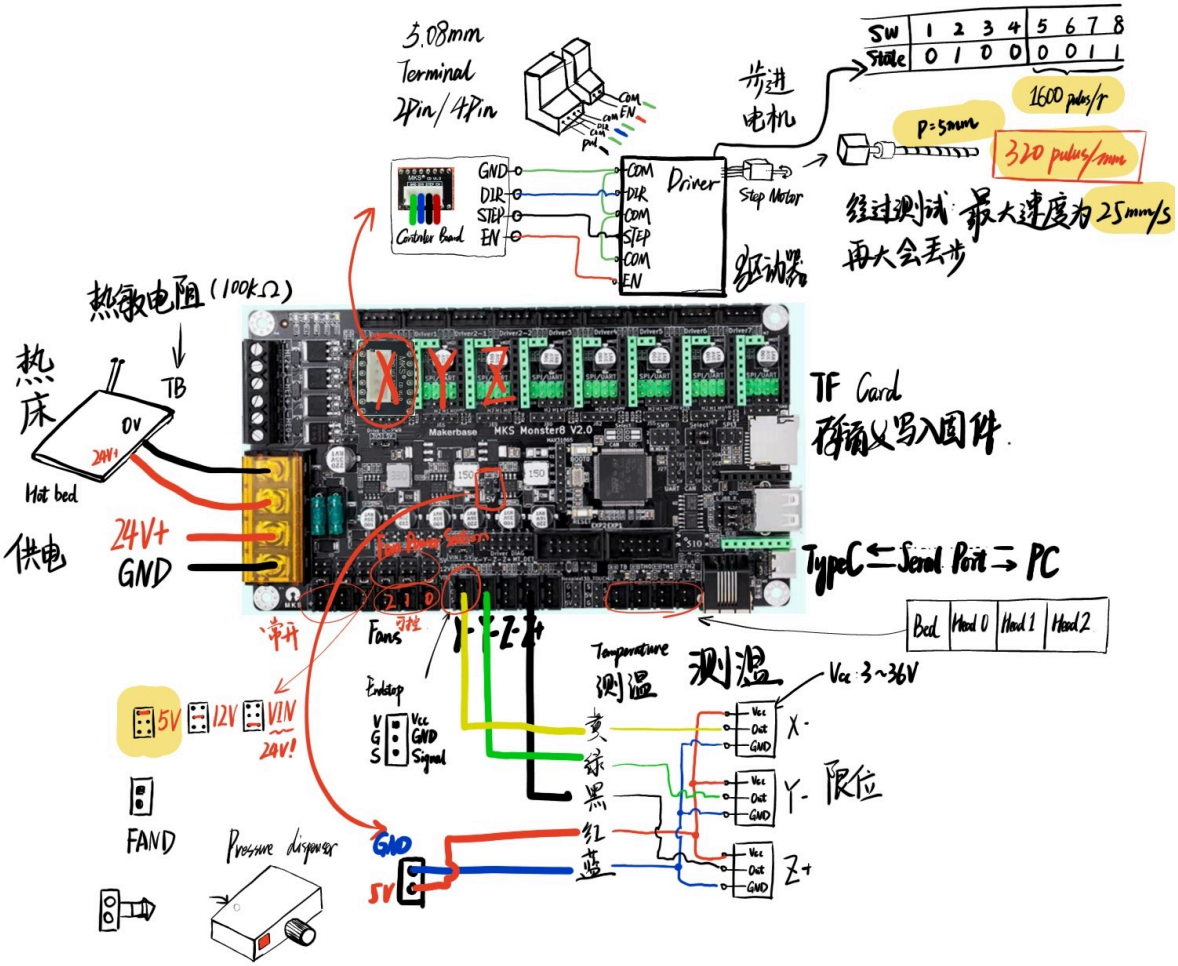
14 Z+ 限位电压选择, 5V or VIN　15 DIAG信号　16 显示屏

17 USB-C接口　1 外接步进电机驱动　2 U盘

Wiring Configuration

Our platform controls X/Y/Z axes only — no extrusion heating is required. The hot bed is reserved for the substrate, and the fan control port is repurposed for air pressure switch control and limit switches.

> **Figure 6.** Control board wiring for DIW platform.



Firmware Configuration

> **Note:** Only modify the firmware if you need to adjust parameters or troubleshoot. The default configuration requires no changes.

Marlin is the most widely used open-source 3D printing firmware. Key customizations for this platform:

1. **G05 spline interpolation** — enabled for smooth motion
2. **XYZ 3-axis configuration** — pulse equivalent set (pulses per mm for each axis)
3. **Cold extrusion protection disabled** — temperature sensor off, minimum extrusion temperature set to 0°C
4. **Build volume** — adjusted to match actual travel range
5. **Homing direction** — set to X-, Y-, Z+

Development Environment: VS Code with PlatformIO extension

Resources:
- Marlin official: https://marlinfw.org/
- Original config: https://github.com/MarlinFirmware/Configurations/tree/release-2.1.2.4
- Custom Marlin build: https://lecloud.lenovo.com/share/2JQMt9CWcRS9uVE4x (password: `pgjg`)

1.3 Stepper Motors, Drivers & Endstops

Stepper Motors

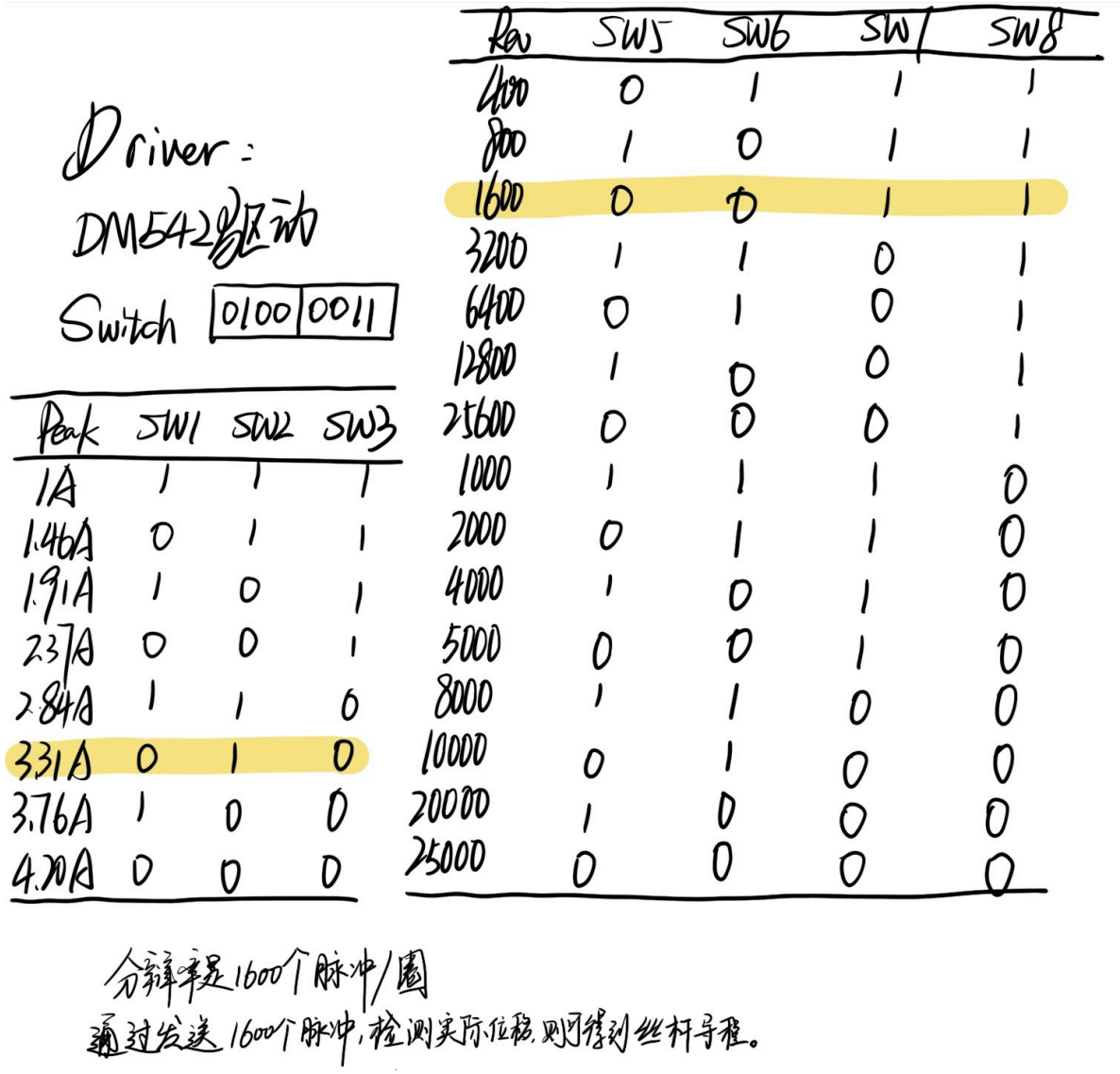
The platform uses **57-series stepper motors** (NEMA-23) for all three axes.

Working Principle: Position control via pulse signals. Each pulse rotates the motor by a specific angle. For a standard 57 stepper:

- **Step angle (θ):** 0.8° (1 pulse = 0.8° rotation)
- **Microstepping:** Driver divides each step by coefficient `n`, effective step angle = θ/n
- **Accuracy note:** Microstepping beyond 32× yields diminishing accuracy returns; higher settings primarily improve smoothness

Driver Configuration

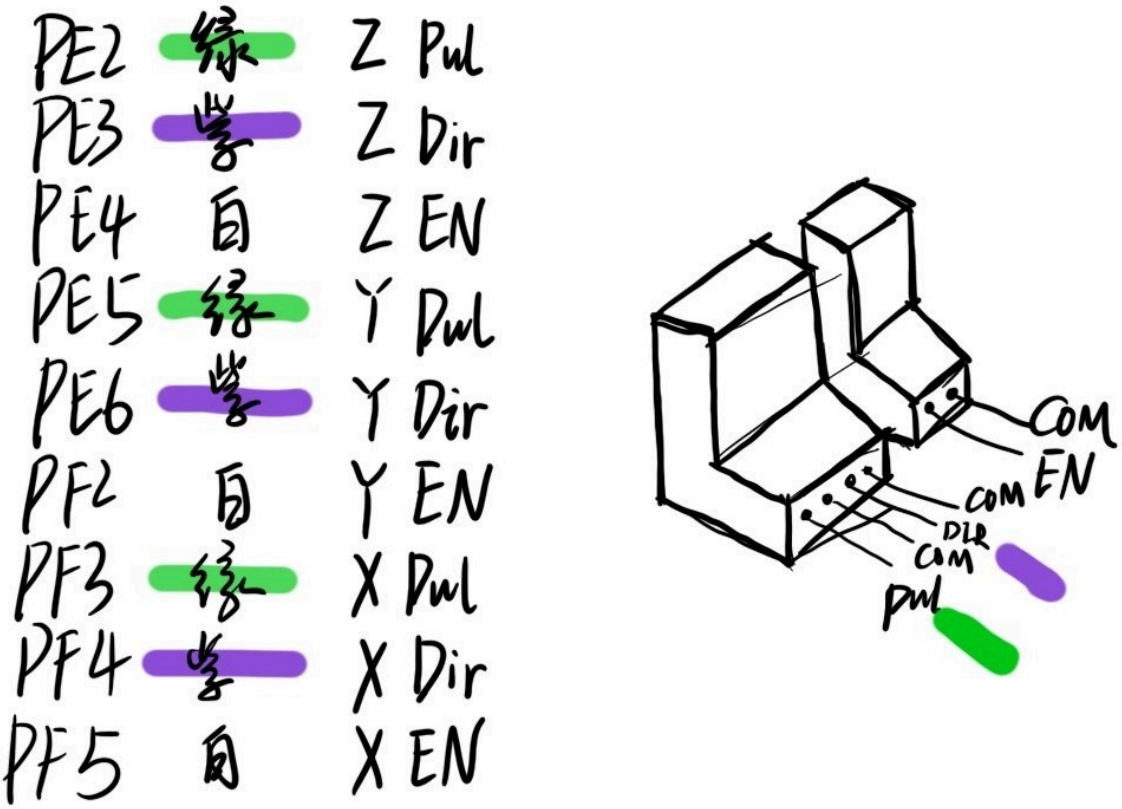
> **Figure 7.** Driver dip switch settings (highlighted = original configuration used).



Current peak value and microstepping resolution are configured via dip switches on the driver.

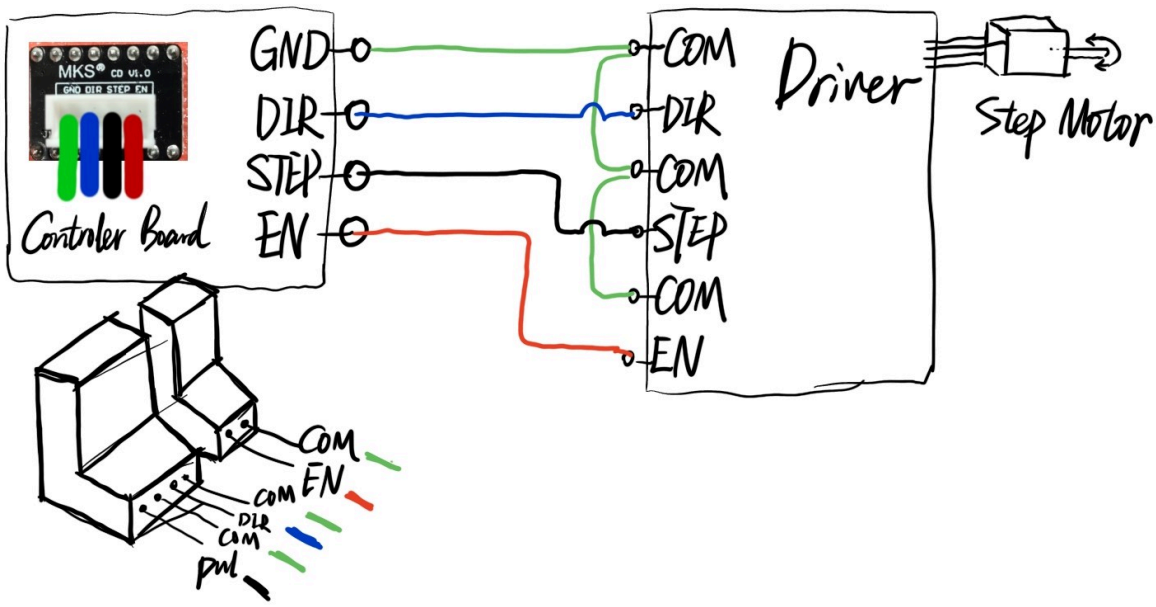
Motor Wiring

> **Figure 8.** Original stepper motor wiring reference.



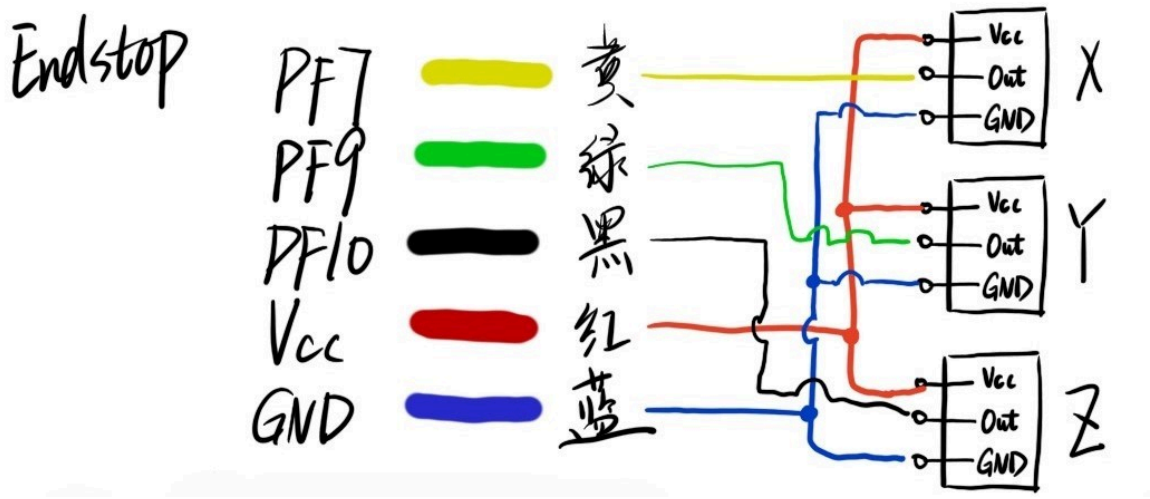
An adapter board interfaces with the controller. Terminals: **COM** (ground), **Enable**, **Pulse**, **Direction**. Wire color coding:

> **Figure 9.** Stepper motor cable color reference.



Endstop Sensors

> **Figure 10.** Endstop wiring configuration.



> ⚠️ **Important:** The default 5V supply proved unreliable during testing. Endstop sensors require **24V** (rated 0-24V). Adjust the wire connection to select the appropriate voltage — the Z+ endstop supports both voltage options via jumper selection.

1.4 Pressure Dispenser

The pneumatic switch is controlled via the fan output port: the controller adjusts the voltage to 24V, triggering a relay module that switches the pressure dispenser on/off.

Technical Specifications

Parameter	Value
Dispensing Modes	16 programmable modes with time display
Dispensing Time	0.01–99.99 s
Interval Time	0.1–9.9 s (auto mode)
Input Voltage	220V±10% 50Hz / 110V±10% 60Hz
Control Voltage	12VDC / 24VDC
Input Pressure	10–100 PSI
Output Pressure	1–80 PSI
Dimensions	235×225×63 mm
Weight	2.8 kg

Mode Configuration

The 4-position DIP switch on the rear panel selects between 16 dispensing modes. Full manual: JND-983A dispensing guide

Mode	S1	S2	S3	S4	Function
1	Off	Off	Off	Off	**Inching** — hold to extrude, release to stop
2	On	Off	On	On	Hold — press once to start timed extrusion, press again to stop
3	Off	On	On	On	Inching timing — hold for timed extrusion
4	On	On	On	On	Auto continuous — extrude according to set time
5	On	Off	Off	Off	Single-shot — 1 dispense per press
6	Off	On	Off	Off	Single-shot — 2 dispenses per press
7	On	On	Off	Off	Single-shot — 3 dispenses per press
8	Off	Off	On	Off	Single-shot — 4 dispenses per press
9	On	Off	On	Off	Single-shot — 5 dispenses per press
10	Off	On	On	Off	Single-shot — 6 dispenses per press
11	On	On	On	Off	Single-shot — 7 dispenses per press
12	Off	Off	Off	On	Single-shot — 8 dispenses per press
13	On	Off	Off	On	Single-shot — 9 dispenses per press
14	Off	On	Off	On	Single-shot — 10 dispenses per press
15	On	On	Off	On	Single-shot — 11 dispenses per press
16	Off	Off	On	On	Single-shot — 12 dispenses per press

> **Note:** This platform uses **Mode 1 (Inching)**.

Operation

1. **Pressure adjustment** — rotate the black knob to set output pressure
2. **Manual trigger** — press the red button on the left side, or use an external foot pedal (modified for program control)

Additional Resources

- Syringe barrel 3D model (SolidWorks)
- Nozzle 3D model (SolidWorks)
- Nozzle data sheet

---

2. Getting Started

> **Prerequisite:** Request compressed air supply from the Lab admin before first use.

Workflow Overview

```
3D Model → PrusaSlicer → G-code Editing → Pronterface → Controller Board
                                                              ↓
                                                              ┴
                                                      ┌───────┴────────┐
                                                      ↓                ↓
                                              XYZ Stepper Motors    Fan Port (M106/M107)
                                              (position control)    → Relay → Pressure valve
```

The PC sends G-code to the controller board's serial buffer. The board decodes and interpolates the instructions into control signals (enable, direction, pulse) for the stepper motor drivers. Meanwhile, `M106` / `M107` commands control the fan port voltage, triggering the relay to switch the pneumatic valve on/off.

2.1 Prepare G-Code

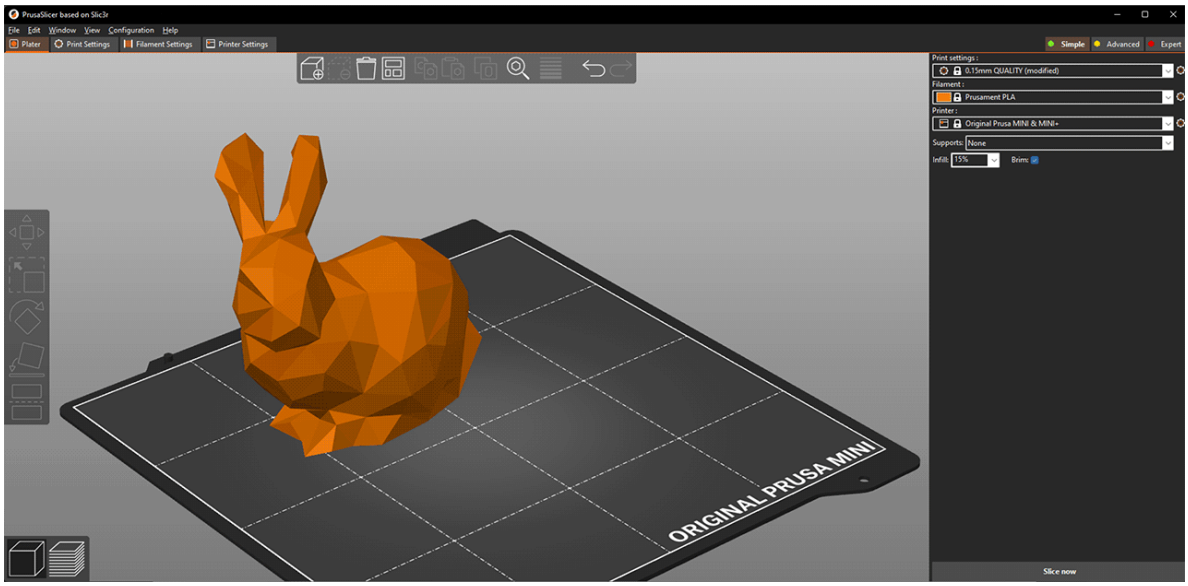
Slicing Software

PrusaSlicer (Recommended)

PrusaSlicer is an actively maintained fork of Slic3r with an active community and fewer bugs.

- **Download:** https://www.prusa3d.com/page/prusaslicer_424/

> **Figure 11.** PrusaSlicer interface.



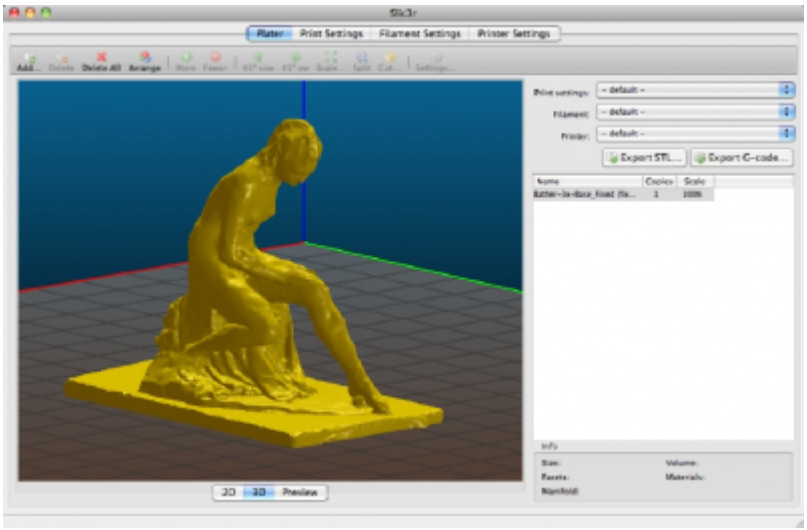
Customized profiles for the AML DIW platform are provided for both **0.4mm** [3] and **0.8mm** [4] nozzles.

Slic3r (Legacy)

> ⚠️ Slic3r is no longer maintained and contains known bugs. Use PrusaSlicer instead.

- **Download:** https://slic3r.org/

Figure 12. Slic3r interface.



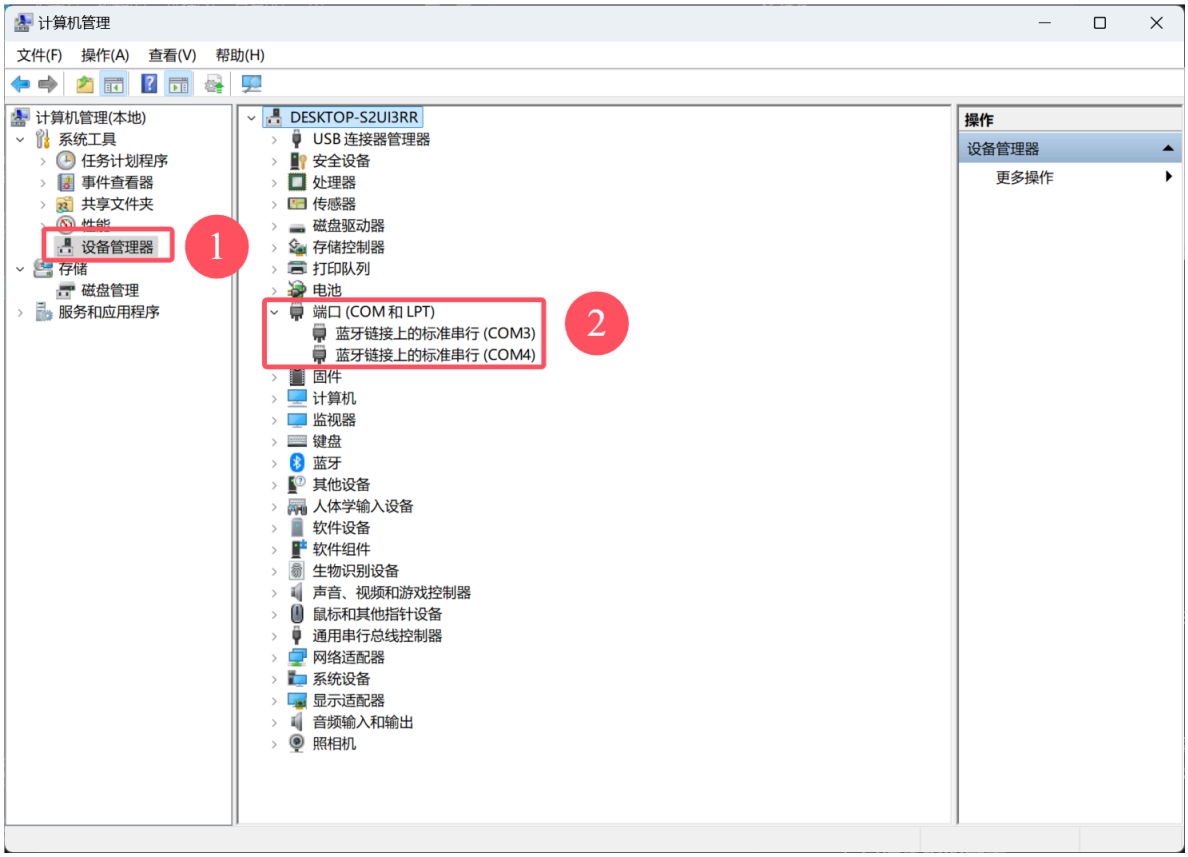
2.2 Connection & Communication

Step 1: Connect Controller to PC

Connect the controller board via USB. Identify the COM port:

- **Windows 10 and earlier:** Right-click "This PC" → Manage → Device Manager → Ports (COM & LPT)
- **Windows 11:** Shift + Right-click "This PC" → Manage → Device Manager → Ports (COM & LPT)

Figure 13. Identifying the COM port in Device Manager.

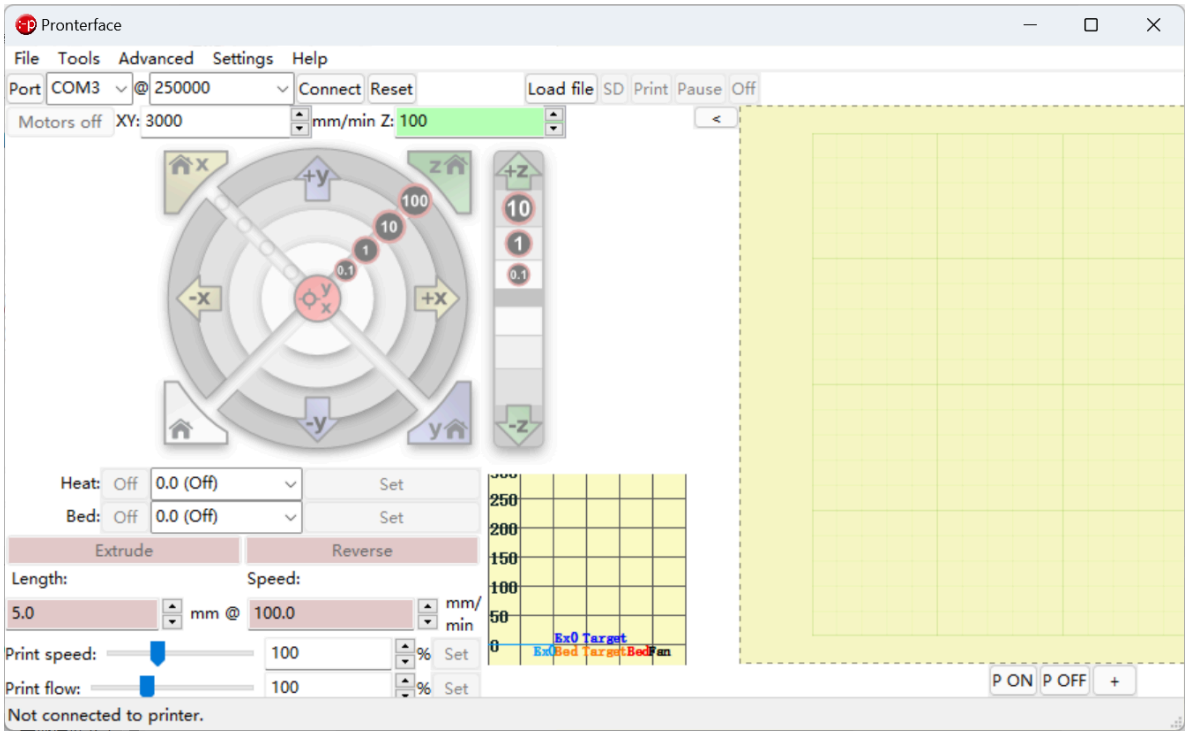


Step 2: Open Pronterface

Pronterface is the host interface for sending G-code to the controller.

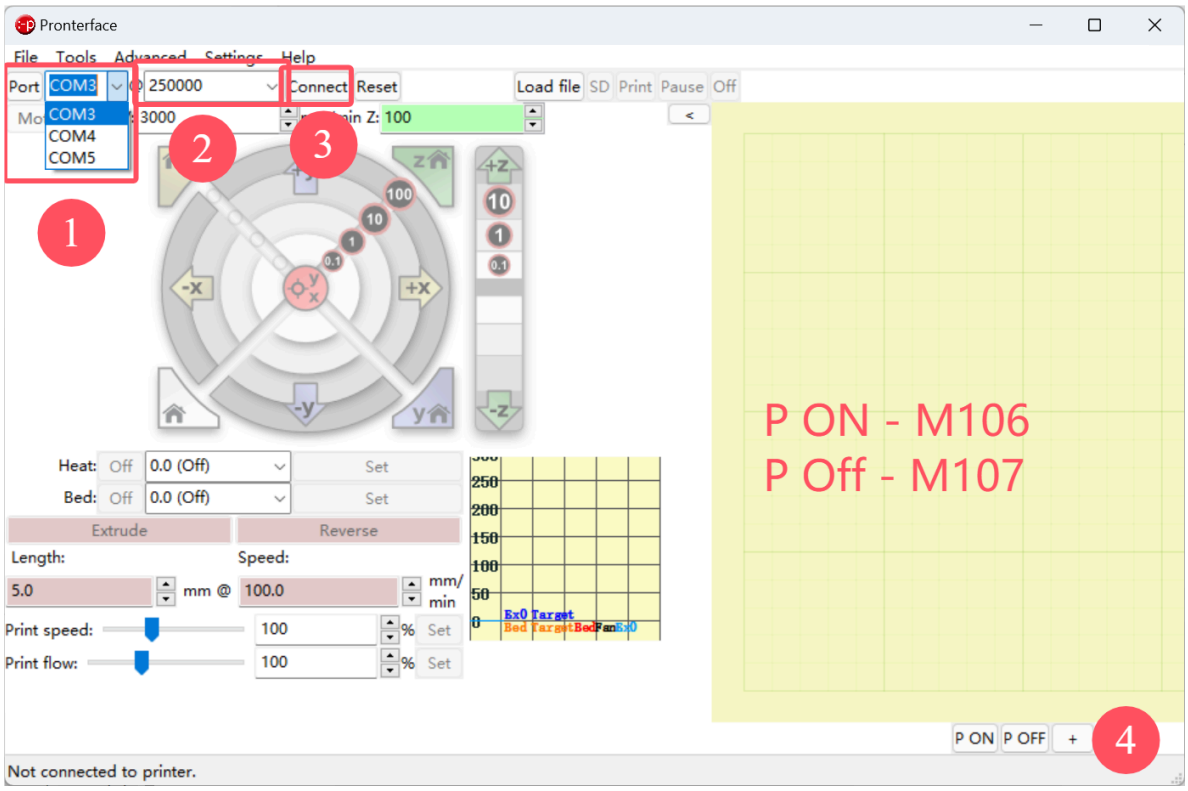
- **Download:** https://www.pronterface.com/

> **Figure 14.** Pronterface interface.



Select the correct COM port and baud rate, then click **Connect**:

> **Figure 15.** Establishing connection in Pronterface.



Verify the connection by manually controlling each axis and the pneumatic switch.

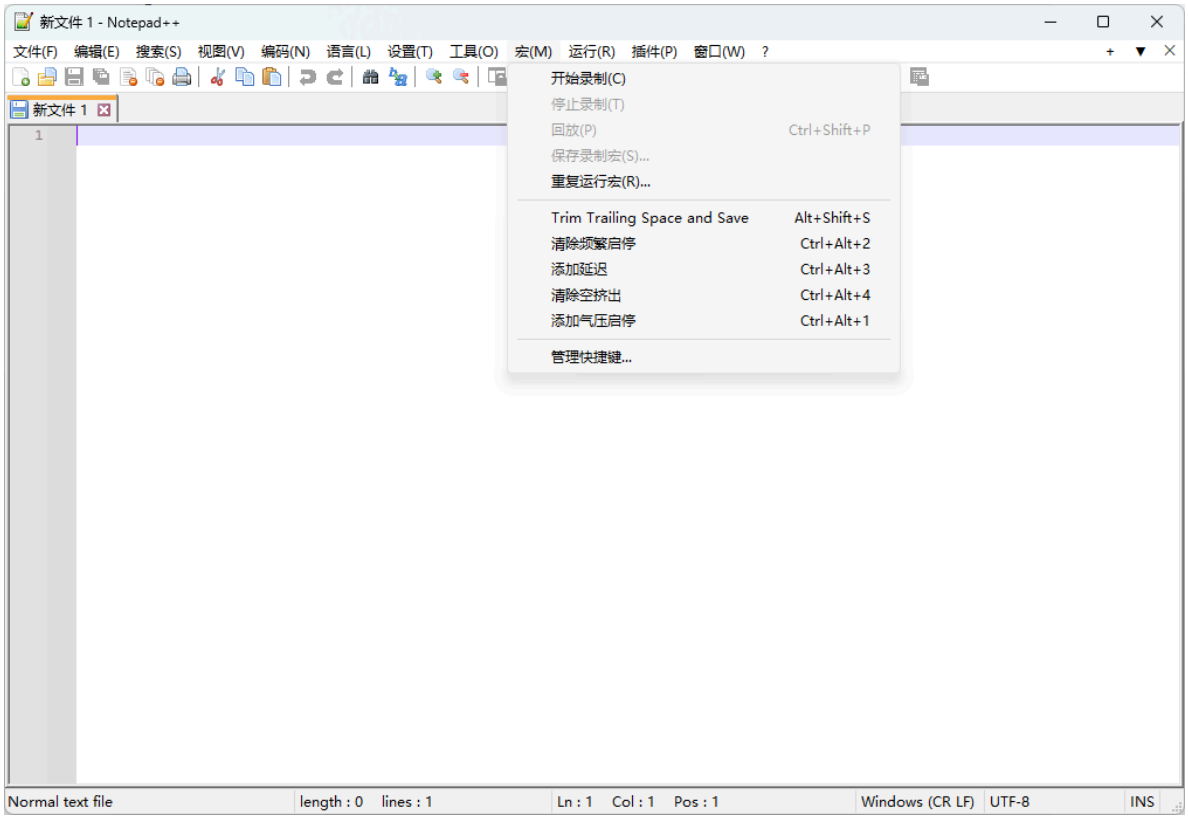
> **Tip:** Marlin G-code reference — https://marlinfw.org/meta/gcode/

3. Tips & Troubleshooting

Editing G-code with Notepad++

Notepad++ supports macro recording, which is invaluable for batch G-code editing.

Figure 16. Macro recording in Notepad++.



Macro Recording Best Practices

1. **Use keyboard shortcuts** — avoid mouse clicks during recording. Use `Home` / `End` instead of arrow keys to move the cursor
2. **Mind the end cursor position** — ensure the final cursor position doesn't cause the macro's search to loop infinitely
3. **Use escape characters** — e.g., replace `M106\nM106` (duplicate lines) with a single `M106` using regex search/replace

Common Issues

Issue	Solution
Endstops not triggering	Verify 24V supply (5V is insufficient)
Inconsistent extrusion	Check compressed air pressure (10-100 PSI input, 1-80 PSI output)
Firmware won't compile	Use VS Code + PlatformIO; verify correct board settings
Pronterface can't connect	Confirm COM port and baud rate match the firmware configuration

Support Materials

#	Resource	Link
1	Pressure dispenser manual	JND-983A 点胶机详细功能设置.docx
2	Syringe barrel 3D model (SLDPRT)	syringe.zip
3	Nozzle 3D model (SLDPRT)	tips.rar
4	Nozzle data sheet	技术表格.pdf
5	Assembly — Z-axis carriage (STEP)	Assembly-Carriage-Z.STEP
6	Printable fixtures (STL)	STLs
7	PrusaSlicer profile — 0.41mm nozzle	Set_Blue.ini
8	PrusaSlicer profile — 0.84mm nozzle	Set_Green.ini

> **Maintained by** NUS CDE AML Lab · **License:** GPL v3 · **Last updated:** 2026-10-16

1. Z-axis carriage assembly (STEP)
2. Printable fixtures (STL)
3. 0.41mm nozzle PrusaSlicer profile
4. 0.84mm nozzle PrusaSlicer profile